

- 6 (a) Define electric field at a point.

.....
 [1]

- (b) An isolated conducting sphere in a vacuum has a capacitance of 69 pF.
 The charge on the sphere is +83 pC.

- (i) On Fig. 6.1, draw field lines to represent the electric field outside the sphere due to the charge on the sphere.

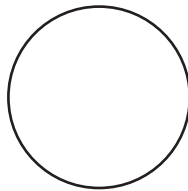


Fig. 6.1

[2]

- (ii) Calculate the electric potential at the surface of the sphere.

electric potential = V [2]

- (iii) Determine the radius of the sphere.

radius = m [2]

